MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 4-ROW RECEPTACLE ASSEMBLIES, RIGHT ANGLE, 100 THROUGH 684 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

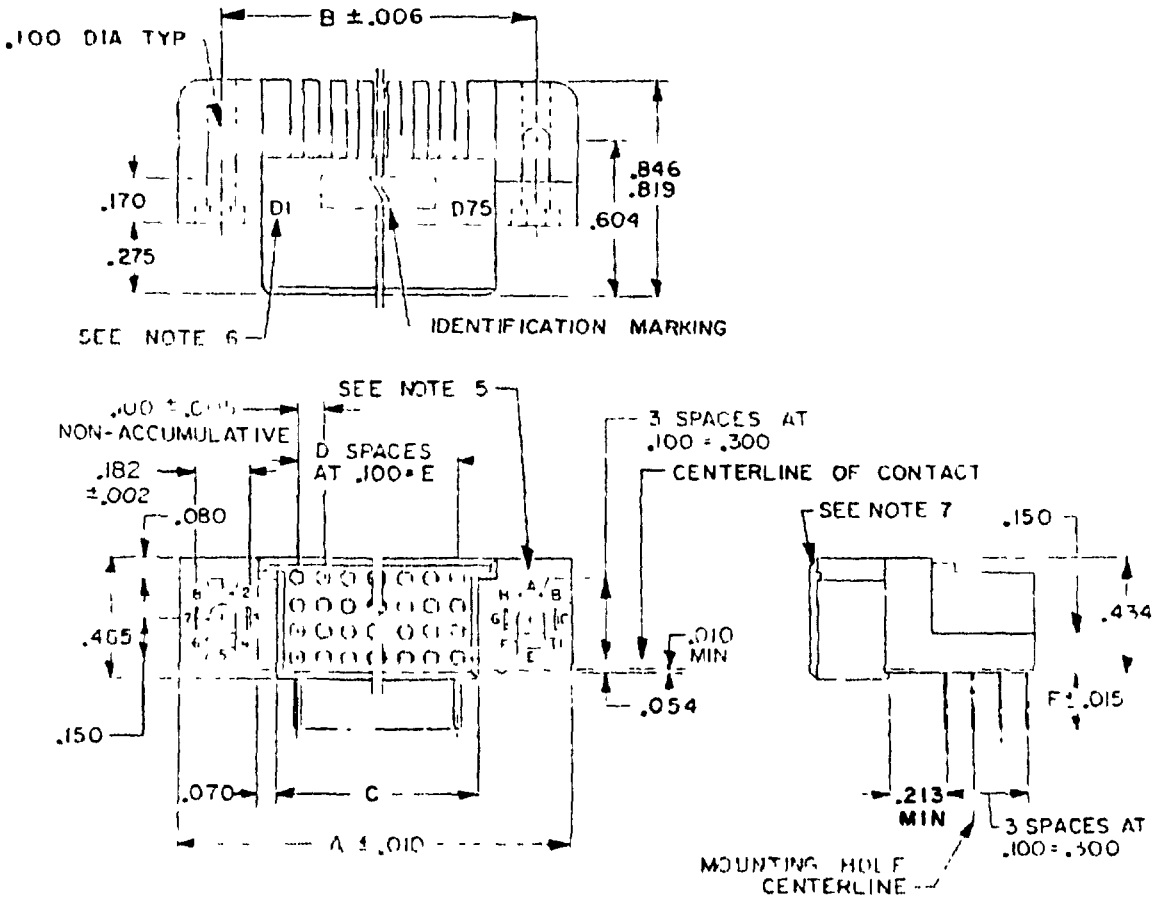
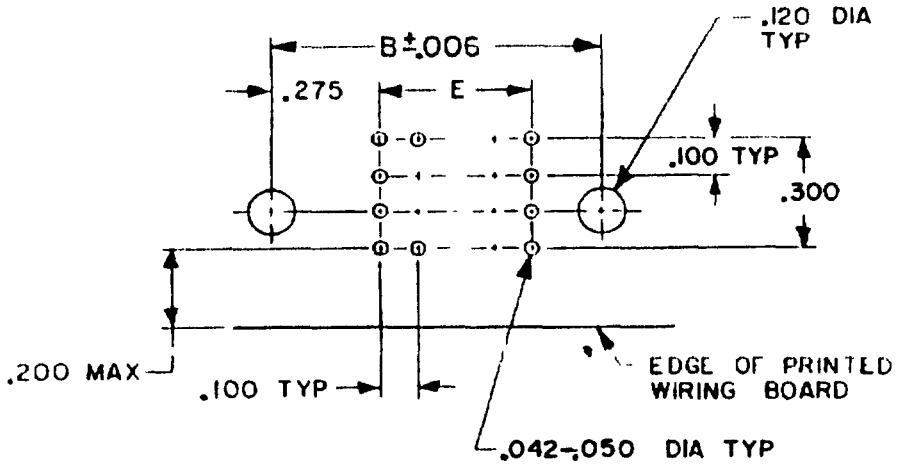


FIGURE 1. Connectors, receptacle assemblies .100 (2.54 mm) spacing, 100 through 300 contacts.

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Recommended circuit layout

F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.180	4 57	I	Solder post (see note 8)
120	3.05	IV	

Inches	mm	Inches	mm
.002	0.05	.150	3.81
.005	0.13	.170	4.32
.006	0.15	182	4 62
010	0 25	200	5.08
015	0 38	.213	5.41
.042	1.07	275	6.98
.050	1.27	.300	7.62
.054	1.37	.434	11.02
.070	1 78	465	11.81
.080	2.03	.604	15.34
100	2.54	819	20.80
120	3.05	.846	21.49

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/179 and /181.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Chamfers shall be .015 (0 38 mm) minimum at a 45° angle ±5°.
8. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0 91 mm) maximum across the diagonal.

FIGURE 1 Connectors, receptacle assemblies .100 (2 54 mm) spacing, 100 through 300 contacts - Continued.

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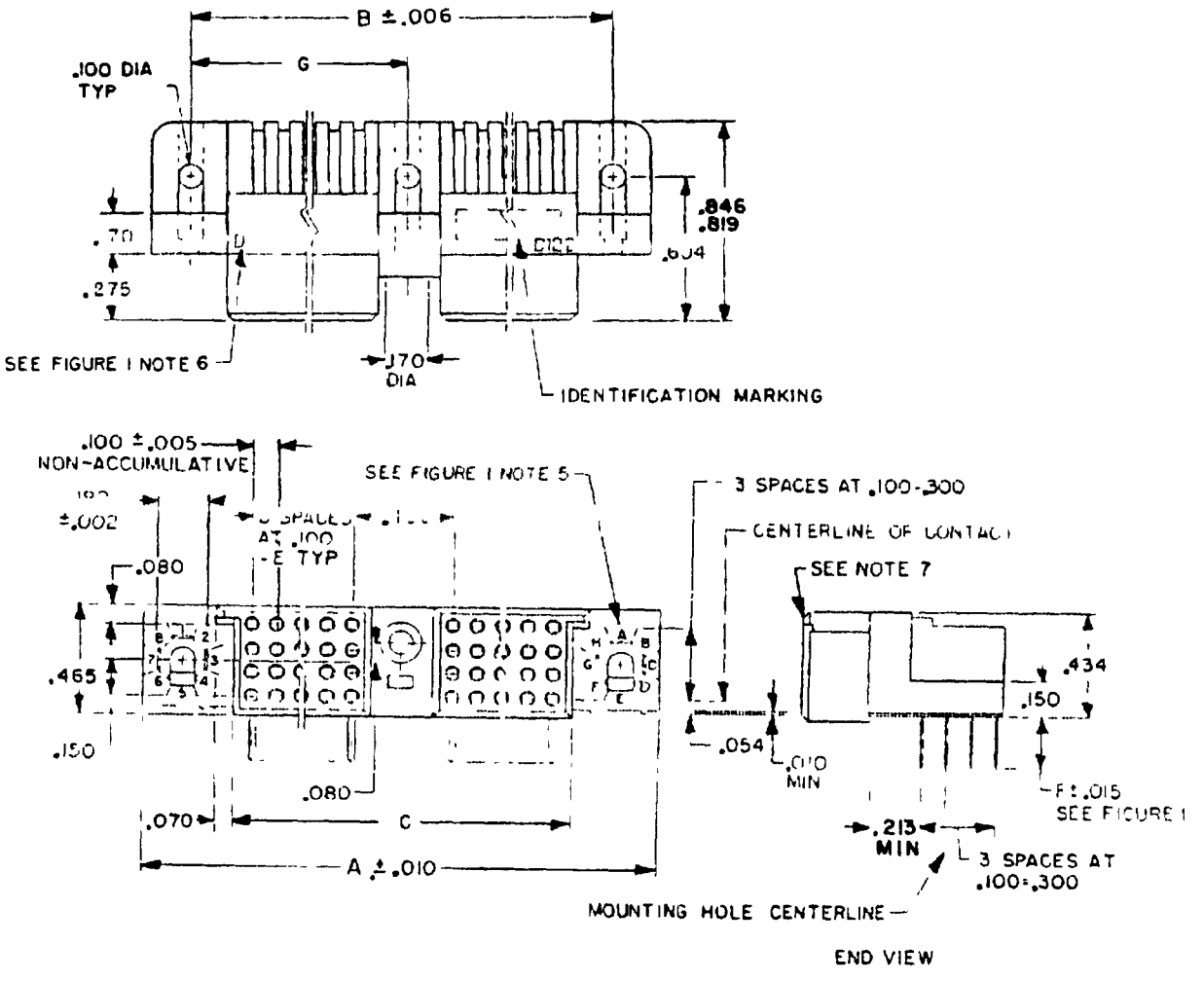
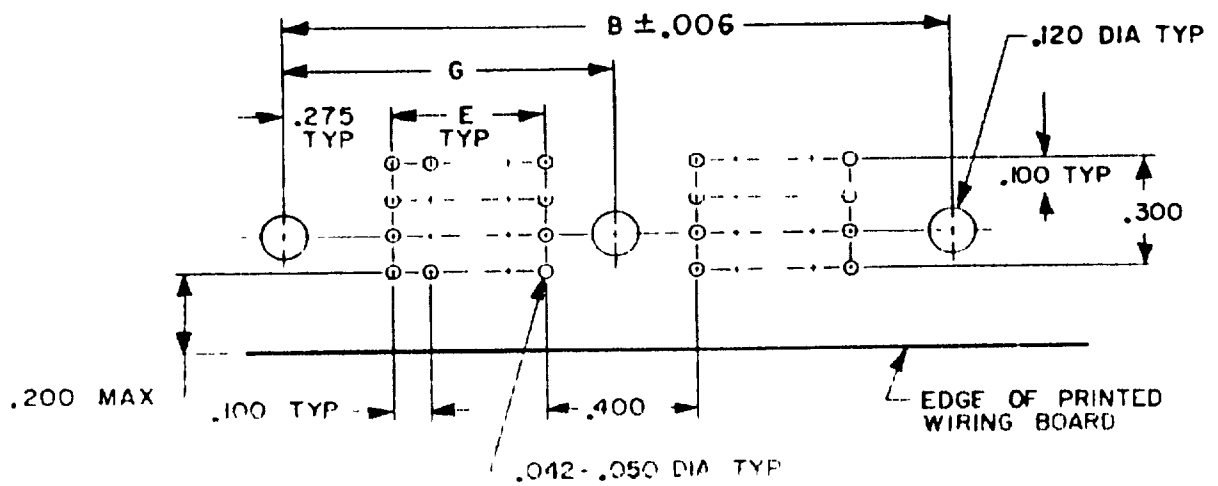


FIGURE 2. Connectors, receptacle assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts.

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Recommended circuit layout

Inches	mm	Inches	mm
.002	0.05	.150	3.81
.005	0.13	.170	4.32
.006	0.15	.182	4.62
.010	0.25	.200	5.08
.015	0.38	.213	5.41
.042	1.07	.275	6.98
.050	1.27	.300	7.62
.054	1 37	434	11.02
070	1 78	.465	11 81
080	2 03	.604	15.34
.100	2.54	.819	20.80
.120	3 05	.846	21 49

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1

FIGURE 2 Connectors, receptacle assemblies .100 (2.54 mm) spacing with one guide, 320 through 480 contacts - Continued.

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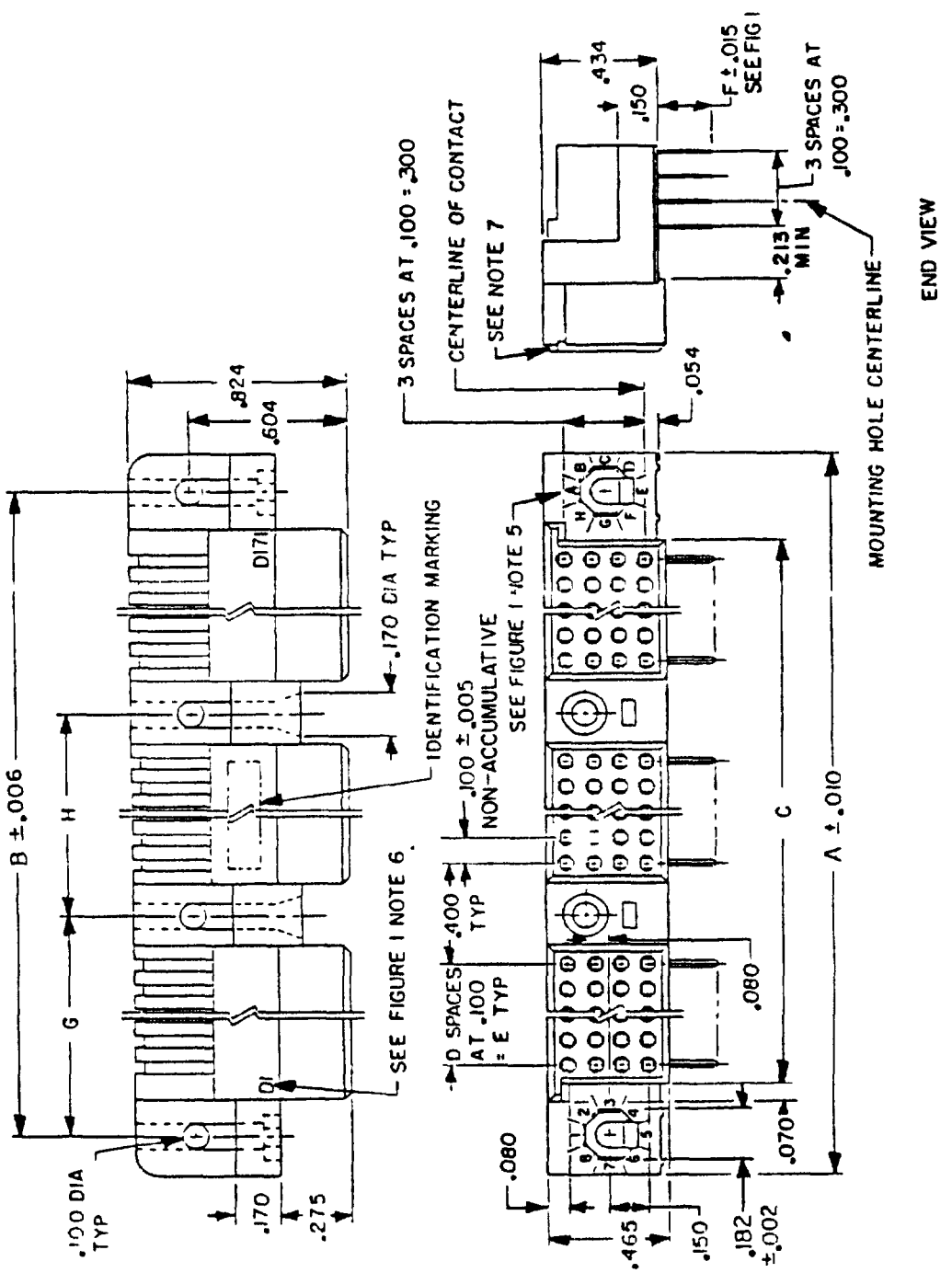
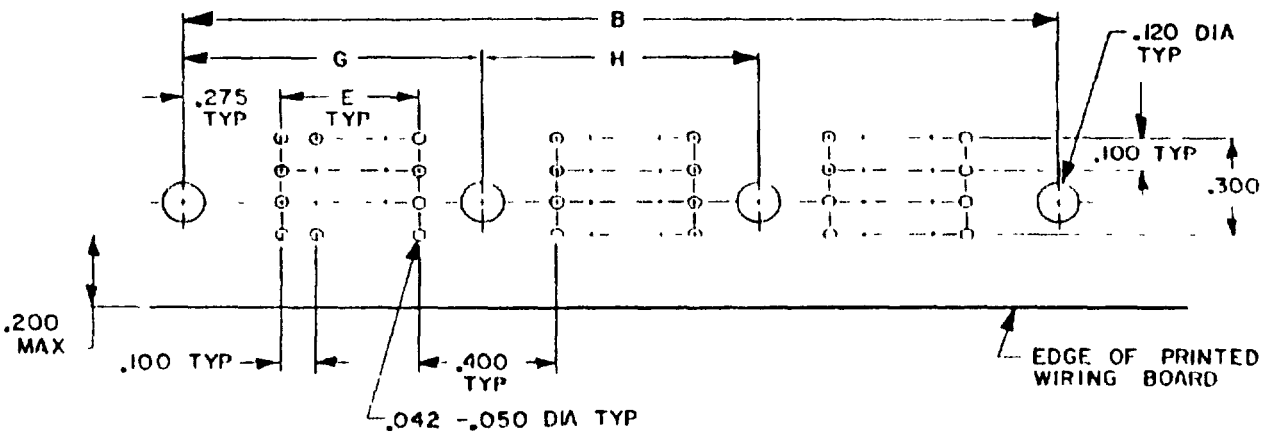


FIGURE 3. Connectors, receptacle assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts.

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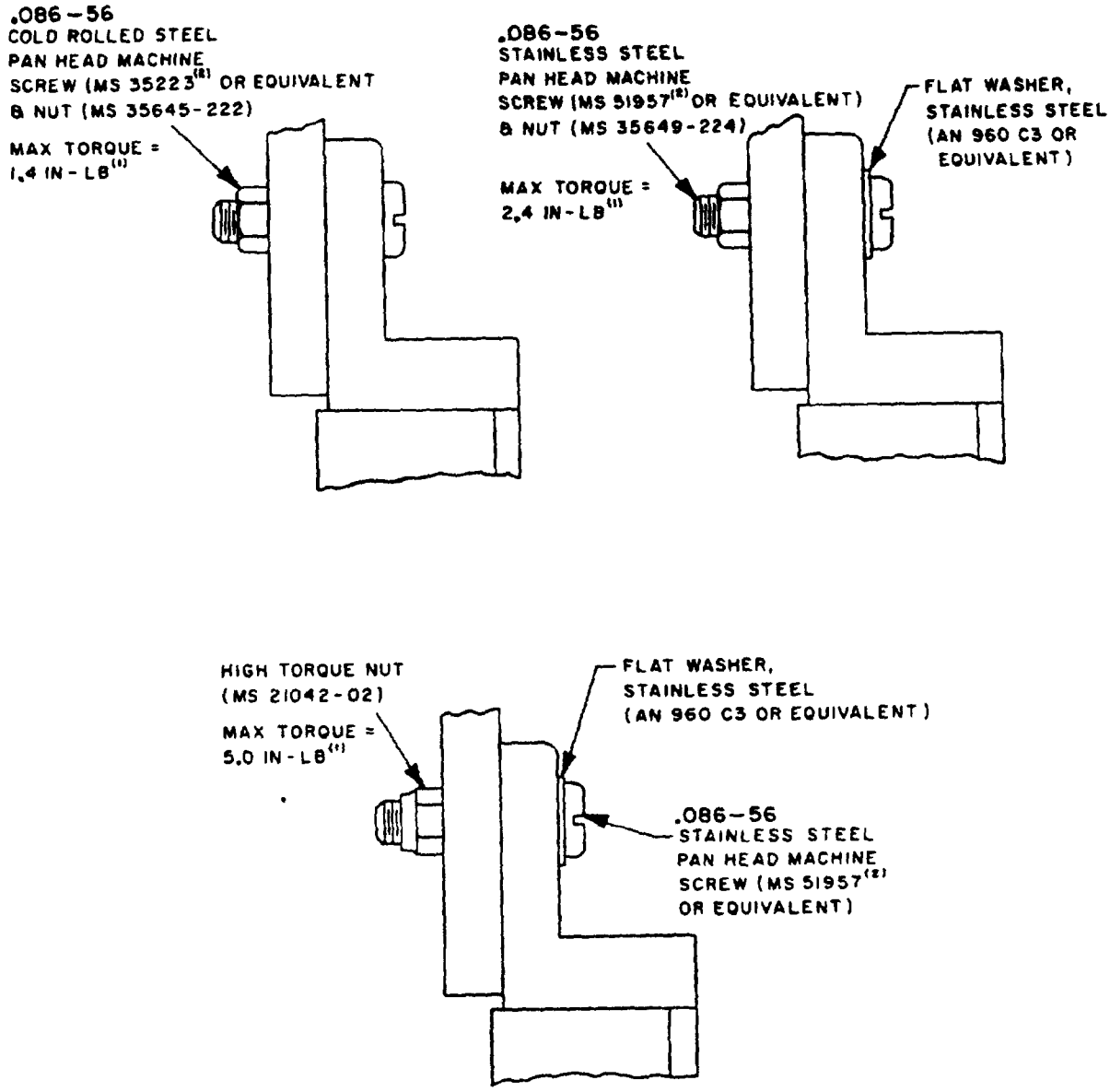
Recommended circuit layout

Inches	mm	Inches	mm
.002	0.05	.150	3.81
.005	0.13	.170	4.32
.006	0.15	.182	4.62
.010	0.25	.200	5.08
.015	0 38	.213	5.41
.042	1 07	.275	6.98
.050	1.27	.300	7.62
.054	1.37	.400	10.16
070	1 78	434	11.02
.080	2.03	.465	11.81
.100	2.54	.604	15.34
.120	3.05	.824	20.93

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With two guides, otherwise same notes as figure 1

FIGURE 3 Connectors, receptacle assemblies 100 (2.54 mm) spacing with two guides, 516 through 684 contacts - Continued.

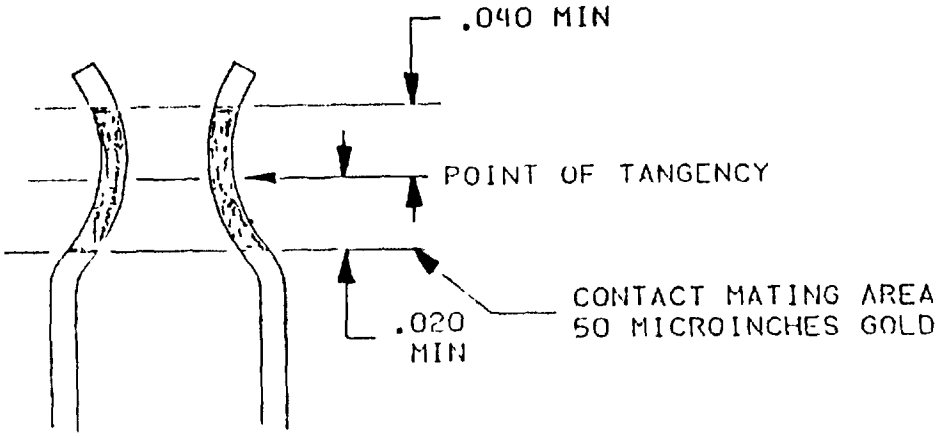
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NOTES
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 4. Mounting recommendations.

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DESIGN OF TINES OPTIONAL

Inches	mm
.020	0.51
.040	1.02

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 5. Contact plating.

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TABLE I Dash number, number of contacts, contact type, and dimensions. 1/

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 04	100	1	I IV	3.250 (82.55)	2.950 (74.93)	2.524 (64.11)	24	2.400 (60.96)			1-25
09 12	128	1	I IV	3.950 (100.33)	3.650 (92.71)	3.224 (81.89)	31	3.100 (78.74)			1-32
17 20	160	1	I IV	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)			1-40
25 28	200	1	I IV	5.750 (146.05)	5.450 (138.43)	5.024 (127.61)	49	4.900 (124.46)			1-50
29 32	220	1	I IV	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)			1-55
33 36	240	1	I IV	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)			1-60
37 40	260	1	I IV	7.250 (184.15)	6.950 (176.53)	6.524 (165.71)	64	6.400 (162.56)			1-65
41 44	268	1	I IV	7.450 (189.23)	7.150 (181.61)	5.724 (170.79)	66	6.600 (167.64)			1-67
45 48	280	1	I IV	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)			1-70
49 52	300	1	I IV	8.250 (209.55)	7.950 (201.93)	7.524 (191.11)	74	7.400 (187.96)			1-75
57 60	344	2	I IV	9.650 (245.11)	9.350 (237.49)	8.924 (226.67)	42	4.200 (106.68)	4.675 (118.75)		1-86
69 72	416	2	I IV	11.450 (290.83)	11.150 (283.21)	10.724 (272.39)	51	5.100 (129.54)	5.575 (141.61)		1-104
81 84	488	2	I IV	13.250 (336.55)	12.950 (328.93)	12.524 (318.11)	60	6.000 (152.40)	6.475 (164.46)		1-222

See footnote at end of table

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued. 1/

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
85 88	516	3	I IV	14.250 (361.95)	13.950 (354.33)	13.524 (343.51)	42	4.200' (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
89 92	540	3	I IV	14.850 (377.19)	14.550 (369.57)	14.124 (358.75)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
101 104	612	3	I IV	16.650 (422.91)	16.350 (415.29)	15.924 (404.47)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
105 108	636	3	I IV	17.250 (438.15)	16.950 (430.53)	16.524 (419.71)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1-159
113 116	684	3	I IV	18.450 (468.63)	18.150 (461.01)	17.724 (450.19)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

1/ Deleted dash numbers in the series 01 through 116 are canceled with no replacement.

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TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish. Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area shall have a minimum plating (see figure 5) in accordance with MIL-C-55032. Solder post area (contact type I, .195 minimum length; type IV, .135 minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.

1/ Type C finish is deleted and superseded by type A or B.

REQUIREMENTS:

Design and construction.

Dimensions and configuration. See figures 1, 2, 3, 4, and table I.

Material In accordance with MIL-C-55302.

Contacts shall be copper-beryllium in accordance with QQ-C-530 or QQ-C-533.

Plating See table II and figure 5.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, Part or Identifying Number (PIN) MS55302/31-04 and two retaining rivets, PIN M55302/31-10 or two .086-56 UNC-2A mounting screws, are recommended. Keys and rivets shall be ordered separately.

Jackscrews (see MIL-C-55302/182)

Use M55302/182-07 or -09. Jackscrews are purchased separately.

Mating and unmating. The maximum mating force in pounds shall be the number of contacts multiplied by .125; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance. No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention. 3 pounds, minimum.

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Dielectric withstanding voltage:

Sea level 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part or Identifying Number (PIN): M55302/180 (and dash number from table I and type from table II).

Example: M55302/180-01A.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians.
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, SH
Air Force - 11, 85, 99
DLA - ES

User activities.
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity.
Air Force - 17

Agent
DLA - ES

(Project 3715-3715-08)

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